

MATHEMATICS

A SYMMETRIC PRIMITIVE NOTION FOR EUCLIDEAN GEOMETRY

BY

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There are several examples known of ternary relations between points that are sufficient as single primitive notions for the various Euclidean geometries. PIERI has shown in [2] that the relation $I(x, y, z)$ (i.e., x is equidistant from y and z) is sufficient for all dimensions greater than 1, while BETH and TARSKI in [1] have demonstrated that for dimensions greater than 2 it suffices to use the relation $E(x, y, z)$ (i.e., the points x, y , and z form an equilateral triangle). The relation E is symmetric but not sufficient for plane geometry. It will be proved in this note that the symmetric relation $S(x, y, z)$ (i.e., the points x, y , and z in some order form a right triangle) can be taken as a single primitive for all dimensions larger than 1. As TARSKI has remarked¹), the relation $H(x, y, z)$ (i.e., the points x, y , and z in some order form an isosceles triangle), which is the symmetrization of PIERI's relation, can also serve as the only primitive notion except in dimension 1.

The following notation will be used:

E_n is the n -dimensional Euclidian vector space of n -tuples of real numbers.

The variables x, y, z, w, u, v shall range over vectors (points).

$x+y, x-y, \alpha x, x \cdot y$ denote respectively the vector sum, difference, scalar product, and inner product of vectors. We set $x^2 = x \cdot x$. O denotes the zero vector.

$C(x, y, z)$ means that the points x, y, z are collinear.

$I(x, y, z)$ can be expressed as $(x-y)^2 = (x-z)^2$.

$M(x, y, z)$ means that x is the midpoint of the segment joining y and z ; in other words $2x = y + z$.

$T(x, y, z)$ is that relation holding between the points x, y, z if they are distinct and form a right triangle with x the vertex at the right angle. For distinct points this relation can be expressed in vector notation by either of the equations

$$(x-y) \cdot (x-z) = 0,$$

or

$$(x-y)^2 + (x-z)^2 = (y-z)^2.$$

¹) In a letter to the author.

$S(x, y, z)$ holds just in case one of $T(x, y, z)$, $T(y, z, x)$, or $T(z, x, y)$ holds.

$S^*(x, y, z)$ holds if either $S(x, y, z)$ holds or the three points are not all distinct. It is obvious that the two relations S and S^* are interdefinable.

Due to the fact that some of our definitions are rather long and involved, logical notation will be used by way of abbreviation. Thus the symbols $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\sim$ denote respectively conjunction, disjunction, implication, equivalence, and negation. $\wedge$ denotes universal quantification, and $\vee$ denotes existential quantification.

By virtue of the result of PIERI it will suffice to show that in all dimensions except 1 the relation I can be defined solely in terms of S with the aid of purely logical notions. The exact definition is effected in several steps as indicated in the following lemmas.

Lemma 1. *For $n \geq 2$ the following equivalence holds in E_n :*

$$I(x, y, z) \leftrightarrow [y = z \vee M(x, y, z) \vee \vee w [M(w, y, z) \wedge T(w, x, y) \wedge T(w, x, z)]]$$

Proof: Assume that $I(x, y, z)$ holds but that $y \neq z$ and that $M(x, y, z)$ does not hold. Thus x, y, z form a non-degenerate isosceles triangle with x at the vertex. Let w be the midpoint of the base. Now w is also the foot of the altitude, and the required condition follows.

Assume next that the right hand side of the equivalence holds. Clearly either of the conditions $y = z$ or $M(x, y, z)$ implies $I(x, y, z)$. Thus we may assume that a point w is chosen so that the formula

$$M(w, y, z) \wedge T(w, x, y) \wedge T(w, x, z)$$

is satisfied. In view of the fact that $M(w, y, z)$ implies $I(w, y, z)$, the above condition yields the three equations

$$\begin{aligned} (w - y)^2 &= (w - z)^2, \\ (w - x)^2 + (w - y)^2 &= (x - y)^2, \text{ and} \\ (w - x)^2 + (w - z)^2 &= (x - z)^2. \end{aligned}$$

These in turn imply

$$(x - y)^2 = (x - z)^2,$$

and hence $I(x, y, z)$. Thus the equivalence holds in both directions.

Lemma 2. *For $n \geq 2$ the following equivalence holds in E_n :*

$$M(x, y, z) \leftrightarrow [[x = y \wedge x = z] \vee \vee u \vee v [T(u, y, z) \wedge T(v, y, z) \wedge T(y, u, v) \wedge T(z, u, v) \wedge T(x, y, u) \wedge T(x, y, v) \wedge T(x, z, u) \wedge T(x, z, v)]]].$$

Proof: Assume first that $M(x, y, z)$ holds. If $y = z$, then it follows that $x = y$ and $x = z$. Assume then that $y \neq z$. In any plane containing x, y , and z , let u and v be the other vertices of a square with the segment $[y, z]$ as a diagonal. Clearly all the combinations of vertices form right triangles as indicated in the first half of the condition, while the last four conjuncts hold because the point x is the intersection of the mutually perpendicular diagonals.

Assume next that the right hand side of the equivalence holds. If $x=y$ and $x=z$, then $M(x, y, z)$ is trivial. Assume then that points u and v are chosen so that the formula

$$\begin{aligned} T(u, y, z) \wedge T(v, y, z) \wedge T(y, u, v) \wedge T(z, u, v) \wedge \\ T(x, y, u) \wedge T(x, y, v) \wedge T(x, z, u) \wedge T(x, z, v) \end{aligned}$$

holds. Without loss of generality we may assume that $x=O$. By reference to the meaning of the relation T the above formula yields at once

$$\begin{aligned} (u-y) \cdot (u-z) &= 0, \\ (v-y) \cdot (v-z) &= 0, \\ (y-u) \cdot (y-v) &= 0, \\ (z-u) \cdot (z-v) &= 0, \\ y \cdot u &= 0, \\ y \cdot v &= 0, \\ z \cdot u &= 0, \text{ and} \\ z \cdot v &= 0. \end{aligned}$$

Whence it follows that

$$\begin{aligned} u^2 + y \cdot z &= 0, \\ v^2 + y \cdot z &= 0, \\ y^2 + u \cdot v &= 0, \text{ and} \\ z^2 + u \cdot v &= 0. \end{aligned}$$

These last equations yield

$$\begin{aligned} (y+z)^2 &= y^2 + 2y \cdot z + z^2 \\ &= -u \cdot v - v^2 - u^2 - u \cdot v \\ &= -(u+v)^2. \end{aligned}$$

But $(y+z)^2 \geq 0$ and $(u+v)^2 \geq 0$. Hence $y+z=O$, and since $x=O$ it follows that $M(x, y, z)$, which completes the proof.

Lemma 3. *The following equivalence holds in E_2 :*

$$C(x, y, z) \leftrightarrow \wedge u \vee v [u \neq v \wedge S^*(u, v, x) \wedge S^*(u, v, y) \wedge S^*(u, v, z)].$$

Proof: Assume first that $C(x, y, z)$ holds and let u be any point. If u is not collinear with x, y , and z , then let v be the foot of the perpendicular from u to the line joining the three points. It is clear that $u \neq v$ and that the indicated relations hold. If on the other hand u is collinear with x, y , and z , then let v be any point on the perpendicular erected at u distinct from u . Again it is clear that the required relations hold, and thus the equivalence holds from left to right.

Assume next that the right side of the equivalence holds but that $C(x, y, z)$ does not hold. Thus x, y , and z form a non-degenerate triangle. Let u be any interior point of this triangle; for example we could let u be the intersection of the medians. Now given any point v distinct from u

the locus $\mathcal{S}^*(u, v)$ of points w such that $S^*(u, v, w)$ holds consists of the points lying either on the circle with the segment $[u, v]$ as diameter or on one of the two tangents to the circle at the ends of this diameter (see figure 1). Any line through the point u divides the plane in such a

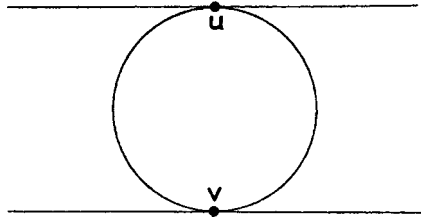


Fig. 1

way that at least two of the vertices of the triangle x, y, z lie on opposite sides of the line. Notice, however, that all the points on the locus $\mathcal{S}^*(u, v)$ lie either on or on one side of a certain line through u . Thus there does not exist a point v distinct from u such that $\mathcal{S}^*(u, v)$ contains all three of the points x, y and z . This contradicts our assumption that the right hand side of the equivalence holds, and hence we must conclude that the three points are collinear.

Lemma 4. *The following equivalence holds in E_2 :*

$$T(x, y, z) \leftrightarrow [x \neq y \wedge x \neq z \wedge y \neq z \wedge \wedge u [C(x, y, u) \rightarrow S^*(x, z, u)]].$$

Proof: The implication from left to right is obvious. Assume then that the right side of the equivalence holds. If we let $u=y$, it follows that $S^*(x, z, y)$ holds and indeed that $S(x, y, z)$ holds. Thus x, y , and z form a non-degenerate right triangle. Assume that y is the vertex at the right angle. The line joining x and y can only intersect the locus

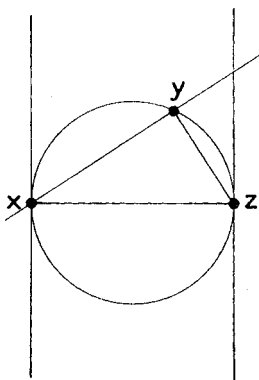


Fig. 2

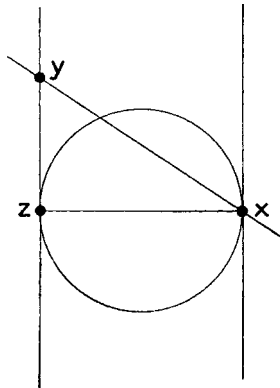


Fig. 3

$\mathcal{S}^*(x, z)$ in three points (see Figure 2), which contradicts the above condition. Similarly the point z cannot be the vertex at the right angle (see Figure 3). Thus the relation $T(x, y, z)$ is satisfied, which completes the proof.

Lemma 5. For $n \geq 3$ the following equivalence holds in E_n :

$$\begin{aligned} T(x, y, z) &\leftrightarrow \forall w [S(x, y, z) \wedge S(x, y, w) \wedge S(x, z, w) \wedge S(y, z, w) \wedge \\ &\sim \forall u [S(x, w, u) \wedge S(x, y, u) \wedge S(x, z, u) \wedge S(y, z, u)]]]. \end{aligned}$$

Proof: We distinguish first the case when $n=3$. In this dimension the locus $\mathcal{S}(x, y)$ of points z such that $S(x, y, z)$ holds consists of those points distinct from x and y lying either on the sphere with the segment $[x, y]$ as diameter or on one of the two tangent planes to the sphere at the ends of this diameter. Assume that the points x, y, z in some order form a right triangle. The intersection $\mathcal{S}(x, y) \cap \mathcal{S}(x, z) \cap \mathcal{S}(y, z)$ consists of those points distinct from x, y , and z lying on one of the two lines perpendicular to the plane of the triangle at either ends of the hypotenuse together with an isolated point w which is the fourth vertex

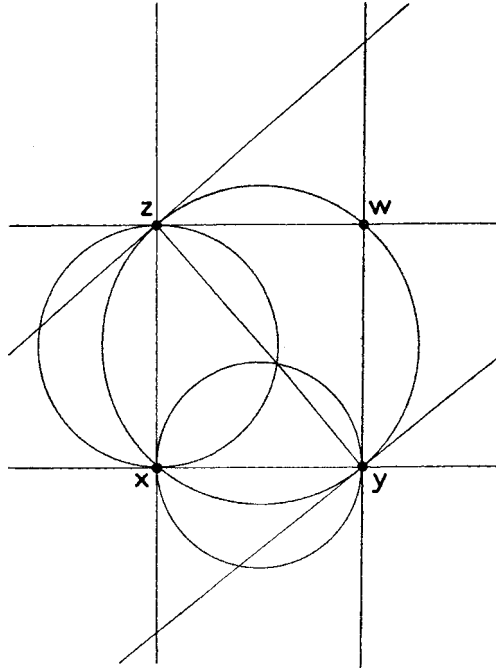


Fig. 4

of the rectangle with x, y , and z as the other vertices (see Figure 4). The point w is distinguished as being that point on the intersection of the loci such that there is no other point on the intersection forming a right triangle with w and the vertex at the right angle of the given triangle. This condition also distinguishes the vertex lying at the right angle of the triangle; whence we derive the definition of T in terms of S .

Consider next the case where $n \geq 3$. Assume first that $T(x, y, z)$ holds. In the plane containing x, y , and z , let w be the fourth vertex of the rectangle with x, y , and z as the other vertices. It is obvious that the

first four conjuncts of the right hand side of the equivalence are satisfied. Suppose that there is a point u satisfying the formula

$$S(x, w, u) \wedge S(x, y, u) \wedge S(x, z, u) \wedge S(y, z, u).$$

The points x, y, z , and w are coplaner, and hence all five of the points are contained in some three dimensional subspace²⁾ of E_n . By the first part of our argument it is impossible that the five points satisfy all these conditions in the subspace. This contradiction shows that the equivalence holds from left to right. The implication in the other direction can be established by a very similar argument using a relativization to a suitable three dimensional subspace. The proof of Lemma 5 is thus complete.

By virtue of the result of PIERI the above lemmas yield the

Theorem. The symmetric ternary relation holding for points which form right triangles can be used as a single primitive notion for Euclidean geometry in any dimension greater than one.

A curious aspect of the proof presented above is the distinction that seemed necessary between the cases of dimension 2 and dimension 3³⁾. The answer to the following question does not seem obvious to the author: *if a notion is sufficient for plane geometry, is it also sufficient for all higher dimensions?* The permissible notions we consider in particular are finitary relations between points that are meaningful in all dimensions and that retain their proper meanings when relativised to subspaces, as was the case with all relations considered in this paper.

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²⁾ The word *subspace* is understood in the geometric sense and not in the sense of linear vector spaces.

³⁾ The same peculiarity was noticed by TARSKI with regard to the relation H mentioned above.